

SQL Data Warehouse Analytics Project

Project Overview

The **SQL Data Warehouse Analytics** project is an end-to-end data warehousing and analytics solution developed using **Microsoft SQL Server** and **T-SQL**. The project demonstrates how raw business data can be transformed into meaningful insights through dimensional modeling, efficient data loading, and analytical reporting.

The primary objective of this project is to design a structured data warehouse, load transactional data from CSV files, and perform SQL analysis to generate business insights.

Objectives

- Design a data warehouse using a **Star Schema**.
- Import structured data from CSV files into SQL Server.
- Perform exploratory data analysis (EDA).
- Generate business KPIs and performance reports.
- Analyze customer purchasing behavior.
- Analyze product performance and sales trends.
- Demonstrate advanced T-SQL concepts for analytical reporting.

Technology Stack

Technology	Purpose
Microsoft SQL Server	Database Management System
T-SQL	Database programming and analytics
SQL Server Management Studio (SSMS)	Query execution and database management
CSV Files	Source data

Project Architecture

The project follows a **Star Schema** consisting of one fact table and two dimension tables.

Schema

gold

Dimension Tables

- gold.dim_customers
- gold.dim_products

Fact Table

- gold.fact_sales

Database Design

Customer Dimension

Stores customer information including:

- Customer ID
- Customer Number
- First Name
- Last Name
- Country
- Gender
- Marital Status
- Birth Date

Product Dimension

Stores product information including:

- Product ID
- Product Name
- Category
- Subcategory
- Product Line
- Cost
- Maintenance
- Start Date

Sales Fact Table

Stores transactional sales information including:

- Order Number
- Customer Key
- Product Key
- Order Date
- Shipping Date
- Due Date
- Sales Amount
- Quantity
- Price

Data Loading

The datasets are imported into SQL Server using the **BULK INSERT** command.

Example:

```
BULK INSERT gold.fact_sales
FROM 'path_to_csv_file'
WITH (
    FIRSTROW = 2,
    FIELDTERMINATOR = ',',
    TABLOCK
);
```

Project Features

The project includes the following analytical capabilities:

- Database creation
- Schema creation
- Table creation
- Data loading using BULK INSERT
- Exploratory Data Analysis (EDA)
- Business KPI reporting
- Customer segmentation

- Product performance analysis
- Sales trend analysis
- Revenue analysis
- Customer lifetime analysis
- Ranking analysis
- Time-series analysis
- Customer reporting

SQL Concepts Used

Data Definition Language (DDL)

- CREATE DATABASE
- CREATE SCHEMA
- CREATE TABLE
- CREATE VIEW

Data Loading

- BULK INSERT
- TRUNCATE TABLE

Querying

- SELECT
- WHERE
- GROUP BY
- ORDER BY
- DISTINCT

Joins

- LEFT JOIN

Aggregate Functions

- SUM()
- AVG()
- COUNT()
- MIN()
- MAX()

Date Functions

- YEAR()
- GETDATE()
- DATEDIFF()
- DATETRUNC()

Conditional Logic

- CASE

Window Functions

- ROW_NUMBER()
- LAG()
- SUM() OVER()
- AVG() OVER()

Common Table Expressions (CTEs)

- WITH

Business Analysis Performed

The project generates several business reports, including:

- Total sales
- Total customers
- Total products
- Total orders
- Average selling price
- Sales by category
- Sales by country
- Customer revenue
- Top-performing products
- Lowest-performing products
- Customer segmentation
- Customer lifetime value
- Monthly sales trends
- Running total analysis
- Moving average analysis
- Year-over-year product performance

Customer Segmentation

Customers are grouped into three categories:

VIP

- Customer history of at least 12 months
- Total spending greater than 5000

Regular

- Customer history of at least 12 months
- Total spending of 5000 or less

New

- Customer history less than 12 months

Key Skills Demonstrated

- SQL
- T-SQL
- Data Warehousing
- Star Schema Modeling
- Database Design
- ETL (Data Loading)
- Business Intelligence
- Data Analysis
- Window Functions
- Common Table Expressions (CTEs)
- Views
- Reporting
- KPI Development

Project Folder Structure

```
sql-data-warehouse-analytics/  
|  
|— datasets/  
|   |— gold.dim_customers.csv  
|   |— gold.dim_products.csv  
|   └─ gold.fact_sales.csv  
|— scripts/  
|   └─ sqlanalytics_projects  
|— screenshots/  
|  
|— docs/  
|   └─ project_overview.pdf  
└─ README.md
```

Learning Outcomes

Through this project, the following practical skills were developed:

- Designing a dimensional data warehouse.
- Importing external datasets into SQL Server.
- Writing optimized analytical SQL queries.
- Using advanced T-SQL functions for reporting.
- Building customer and sales analytics reports.
- Applying window functions and CTEs to solve business problems.
- Transforming raw transactional data into actionable business insights.

Conclusion

This project demonstrates a complete SQL-based analytics workflow using Microsoft SQL Server. It showcases practical experience in database design, dimensional modeling, data loading, business reporting, and advanced T-SQL programming. The implementation reflects real-world data analytics practices and highlights the ability to convert transactional data into meaningful insights that support business decision-making.